

Haoji(Paul) Bian

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EDUCATION

Northwestern University

Evanston, IL

- Master of Science in Electrical and Computer Engineering
- GPA: 4.00 / 4.00 (top 1%),

Sep 2024 - Dec 2025(Estimate)

University of Wisconsin-Madison

Madison, WI

- Bachelor of Science in Computer Science & Data Science
- GPA: 3.96 / 4.00 (top 2%), Dean's List for 5 Semesters, Summa Cum Laude

Jan 2021 - Dec 2023

PROFESSIONAL EXPERIENCES

HyAxiom

East Hartford, CT

Full Stack Software Developer Intern

June 2025 – Aug 2025

- Use **Java, Spring Boot, PostgreSQL & AWS** to develop industrial energy systems, collaborated with cross-functional engineering teams (electrical, systems, mechanical) to integrate software with hardware subsystems
- Bridge **Salesforce** and remote monitoring systems, develop authentication middleware with **Microservice Lambda, REST API, DynamoDB & Cognito**, allowing for role-based control with fine-grained permissions
- Use **SQS** to queue and consume up to **10,000 QPS** with an asynchronous operation approach at peak times
- Utilize **Redis** to cache **K-V** pattern data under concurrent modifications, reducing the pressure on the database
- Automate deployment of **serverless** projects w/ **Cloudfront**, simplified the process used to be done manually
- Develop a real-time monitoring and progress tracking system using **WebSocket (WSS)** for efficient emergency event detection, alarming users immediately and enhancing work order and case process

Katfun LLC

Chicago, IL

Full Stack Software Engineer | Project Manager

March 2023 – Sep 2024

- Build and develop "**Katfun.com**", a platform dedicated to enhancing the experience for the pet community
- Design the application using **MERN** and **RESTful** with the **MVC** pattern. Deploy on **AWS EC2** proxied by **Nginx**
- Use **Cache-Aside** Strategy along with **Redis** and **MongoDB** distributed database, to reduce the bottleneck of direct disk IO access during highly concurrent requests and increase data throughput by up to 5x
- Use **Cloudflare CDN & Cloudfront** to proxy with **AWS S3** to solve the problem of insufficient network bandwidth when massive accesses of media resources at the same time. Boost user-end resource loading time by 70%
- Manage the entire project lifecycle, from conceptualization to deployment, to ensure timely delivery and quality assurance, updating 10+ production versions in **Agile** practice using **Git, Scrum** and **Jira** as a Project Manager

One Community Global

San Gabriel, CA

Full Stack Software Engineer

Aug 2023 - June 2024

- Developed responsive front-end interfaces with **React.js**, implementing **JWT**-based stateless authentication with **HTTPS**-only cookies for secure session management, reducing server-side state overhead by 40%
- Optimized **SQL** query and for high-traffic APIs, designed a distributed **key-value** store with leader election and replication, and used Java concurrency utilities frameworks
- Built a distributed web crawler using Python (**BeautifulSoup**) to automate data collection from 50+ target sites, with **Redis**-based deduplication, improving information retrieval efficiency by 60% for market research teams

PROJECT EXPERIENCES

[Zaisland](#) | *Java, Spring Boot, React, Restful API, Redis, OAuth2, S3*

Full Stack Web Developer

April 2023 - Nov 2023

- Developed backend with Spring Boot, integrating Google/Apple OAuth2, Stripe API, and Redis caching
- Used Java-based MVC architecture and applied secure session/token handling
- With **Vue.js**, and strategic application of **search engine optimization** best practices and optimized user interface layouts, the platform improved search engine rankings in key product categories by 40%

SKILLS

- **Programming Languages:** C/C++, Java, SQL, R, Python, TypeScript, JavaScript, C#, PHP
- **Backend:** Serverless, Flask, FastAPI, Node.js, Spring Boot, .NET, REST, Kafka, Redis, PostgreSQL, MongoDB
- **Frontend:** React, React Native, Vue.js, JavaScript, TypeScript, HTML/CSS, Tailwind CSS, AJAX
- **Devops & Tools:** AWS, Azure, Docker, Nginx, Git, Jira, Agile/Scrum, CI/CD, Windows, UNIX, Linux, Postman